# PA194C: a different 194-state PA consistency machine

Research and reproducible verification prepared for Vladimir Reshetnikov

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**Abstract**

We construct an explicit deterministic, binary, single-tape machine with 194 working states whose blank-tape nonhalting is provably equivalent in PA to Con(PA). The construction meets the requested alternative of a materially different 194-state design. No machine with fewer than 194 states is established by this investigation.

PA194C combines the supplied 67-reader/seven-fold PA194 allocation with the Combined command controller previously implemented at 195 states. Six native transition rows change. The resulting source has 3,649 primitive instructions, compared with the existing PA194's 3,673, and its entire resolved instruction graph equals the previously calibrated PA195C source. The count is $134 + 67 - 7 = 194$. Its transition graph is distinct from all supplied 194-state graphs even after arbitrary state renaming, forgetting the initial state, reflecting the tape, and exchanging symbols.

All 4,749 arbitrary-register source outcomes pass for both the raw and final tables. A separate literal-tape checker rederives four generic list lemmas and verifies the final table without arithmetic-box oracles. Exact blank execution reaches the complete source frame after 57,922 transitions. We compare all 36 incoming archives at a pinned repository revision, distinguish supported constructions from failed diagnostics, and record the unsuccessful searches below 194. The proof is ordinary mathematics with executable supporting certificates; a new end-to-end Lean or Rocq theorem is not claimed.

# Contents

**1 Result, conventions, and scope** **3**
1.1 What has and has not been established . . . 3

**2 Comparison of the supplied constructions** **3**
2.1 The three positive 194-state designs . . . 4
2.2 The relevant 195-state families . . . 5
2.3 Larger machines and formalization layers . . . 5

**3 The new literal construction** **5**
3.1 Six native rows and two source nodes . . . 5
3.2 The unchanged decoder quotient . . . 6

**4 The exact numerical calculus retained by the construction** **7**
4.1 Axiom heads and gate . . . 7
4.2 Freshness and the separation of the diagonal head . . . 8
4.3 Logical bridge used in the theorem . . . 8

**5 The changed command controller and its PA calibration** **9**
5.1 Stores, numerical aliases, and the two changed rules . . . 9
5.2 Uniform stack identities for ten pushed entries . . . 10
5.3 Completeness and soundness at the command level . . . 11
5.4 The inherited logical bridge to PA . . . 12
5.5 What transfers from ZFC and concise iPRA research . . . 13

**6 Physical correctness and its verification boundary** **13**

**7 Material difference and exact graph comparison** **14**

**8 The search below 194 states** **15**
8.1 Association, history folds, and the 194-state plateau . . . 15
8.2 Direct runtime searches . . . 16
8.3 A specific rejected scanner shortcut . . . 16
8.4 Additional 194-state seed and remaining directions . . . 16

**9 Reproduction and delivered artifacts** **17**

**A Complete transition table** **18**

**B All 36 incoming archives** **20**

**C Sources and precise dependency boundary** **22**

# 1 Result, conventions, and scope

The definitive table is `construction/tables/pa194c.tm`. It has working states $0, \ldots, 193$, start state 0, and the distinct terminal label 194. Its two-way infinite tape has alphabet $\{0, 1\}$, is initially zero everywhere, and starts with its head at position zero. Every transition writes one symbol and moves exactly one cell left or right. Each working state has exactly two rows, so the table has 388 rows. The halt label has no outgoing rows and is not included in the working-state count.

**Theorem 1.1** (Main result)**.** *Let $H_{M_{194C}}$ be the usual arithmetical sentence asserting that a finite halting trace exists for this exact table from its specified blank start. Then*

$$\boxed{\mathrm{PA} \vdash (\neg H_{M_{194C}} \leftrightarrow \mathrm{Con(PA)}).}$$

The theorem is an assertion of PA-provability and does not assume Con(PA) as an axiom. Actual nonhalting follows if PA is consistent. The source-calibration argument and physical proof are developed below.

The literal dense table has SHA-256

`d759313c03b515293fe06b12769fa1771d3829a2bad89febaa9b4391990f3a4b`.

The native table starts in 74, halts at 254, and has SHA-256

`b35d80b01ce9c2e95c8b29222881f40928247cdeffac322e4643f4f6694cd33c`.

The supplied label list conjugates every row of these two presentations. All 194 working states are reachable in the pointed transition graph. This graph fact does not assert that every state occurs in the actual blank execution. Appendix A prints the complete dense table.

## 1.1 What has and has not been established

The strict target was fewer than 194 states. The successful deliverable is the explicitly permitted same-size alternative. Its novelty is relative to the inspected, pinned corpus, not a claim about every unpublished machine. Neither the failed searches nor the current best construction establish a lower bound of 194.

The proof status has three separate parts. The literal tables, exact source binding, finite local quotients, and symbolic execution certificates are reproducibly checked by the included programs. The logical calibration is an ordinary syntactic proof in the supplied research, with the changed controller's proof compiler exhibited here. The repository's relevant machine-specific formal Lean and iPRA endpoints have additional unfinished obligations. Those endpoints are not used as if they were completed theorems.

# 2 Comparison of the supplied constructions

All comparisons use *ProveIt* revision

`768c2be1a8e4d2fcf8aa3443c7b0f084d145734c`.

The source locations are `Computability/BusyBeaver/Research/PA`, `docs/incoming`, and the ZFC and ConciseIPRA directories named in the request. The 36 incoming ZIP files were bound to their Git blob identifiers before extraction. A recursive census also includes nested archives and 14 direct numeric PA tables. It recognizes complete five-column binary tables in the supported text formats: 3,486 occurrences, 606 distinct byte forms, and 596 distinct labeled transition functions. These counts include ancestors, repeated references, experimental allocations, and failed machines; they are not counts of valid consistency constructions.

The census is a reproducible inventory of recognized explicit tables, not a semantic recognizer for arbitrary programs that could generate a machine. The archive-by-archive document comparison complements it. The only recognized tables below 194 states in this corpus are 6- and 9-state test controls, not PA consistency candidates. All 36 archive contributions are summarized in Appendix B; exact member locators and hashes are in `comparison/census.json`.

## 2.1 The three positive 194-state designs

| Design | Fixed controls | Raw readers | Folds | Startup steps |
|---|---|---|---|---|
| PA194-67 | 134 | 67 | 7 | 58,154 |
| PA194-67-AD | 134 | 67 | 7 | 58,122 |
| PA194-68 | 134 | 68 | 8 | 58,154 |
| **PA194C (this work)** | **134** | **67** | **7** | **57,922** |

The unnumbered `pa194_conpa_2026-11-13.zip` contains the first two tables, with dense hashes beginning `4bfea62b` and `27613aa9`. The archive whose name contains `(2)` contains PA194-68, beginning `9087d70f`. The names of historical workspace copies were not used to identify these inputs; current Git-bound archive contents determine the association.

The two principal PA194 sources have exactly the same complete ordinal instruction graph: 3,673 primitive sites and 4,780 branch outcomes. Each uses the separate 190-token axiom emitter and the original command controller. Their savings are different implementations of the decoder: 67 readers with seven compatible folds versus 68 readers with eight. Their fold counts cannot simply be added because the folds depend on different trees, helper rows, and incoming histories. The A/D companion permutes physical register positions and was already supplied.

We freshly replayed the full supporting physical checks for all three positive tables. The two 67-reader versions additionally passed the independent whole-tape checker, including rederivation of four generic list lemmas. The 68-reader table passed all raw and folded source outcomes and its 32 simultaneous local quotient cases. A fourth supplied 194-state graph is explicitly rejected: its scanner confuses a legal long-zero address with the header. Its local fold certificate does not repair the failed source simulation.

### 2.2 The relevant 195-state families

| Family | Main change | Source sites | States |
|---|---|---|---|
| Original PA195 | Bundled diagonal gate, original controller | 3,700 | 195 |
| PA195S | Separate diagonal head, original controller | 3,673 | 195 |
| PA195C | Separate diagonal head, Combined controller | 3,649 | 195 |
| PA195DC | Bundled diagonal gate, Combined controller | 3,676 | 195 |
| M195S8 | Bundled diagonal source, longer scanner, eight folds | 3,700 | 195 |
| Supplementary separate195 | Another allocation of separate source | 3,673 | 195 |

The useful combination is now visible. PA194-67 provides the 67-reader, seven-fold allocation. PA195C provides a shorter, already calibrated controller using the same separate emitter. Their changes act on compatible parts of the implementation, but compatibility is proved by fresh full-source and full-tape checks rather than presumed from disjoint patch lists.

The PA195DC packet is also distinct: its 191-token bundled gate pushes a different stack interface. Its 4,784 branch outcomes cannot be substituted for the 4,749 outcomes of the present separate-head controller. Likewise, the successful eight-zero scanner in M195S8 pays an extra raw state to make its source addresses legal. Counting its folds without counting that scanner gives a false saving.

### 2.3 Larger machines and formalization layers

The U208 family develops contextual and finite-history folds, source reassociation, and exact physical return interfaces. Several early archives repeat the same two U208 graphs; later helper-layout work contains additional graphs. The James Chang PA196 design replaces much of the larger axiom implementation with a compact numerical source. Subsequent 196-state investigations vary its axiom gate, parameter choices, register positions, command controller, source association, and return scanner. These provide the ingredients used by the positive 195 and 194 families.

Two comparisons prevent misleading claims. A shorter expanded source does not automatically use fewer physical states: the supplied 3,441-site short-negation implementation still has 197 states. Conversely, a larger reader tree can be competitive if it supports more compatible folds, as PA194-68 demonstrates. The relevant objective is the completed physical allocation with every interface checked.

The direct PA repository also contains many proof layers for the original 195-state quotient and arithmetic endpoints. They are not independent machine designs. At the pinned revision, the PA README and `pa194/ipra/README.md` explicitly retain both machine-specific iPRA reductions as assumptions. An ordinary PA proof, a standard-model simulation theorem, and an internally proved iPRA equivalence are distinct claims. This work asserts the first and supplies executable physical evidence; it does not silently promote the conditional formal endpoint.

## 3 The new literal construction

### 3.1 Six native rows and two source nodes

Starting from the supplied PA194-67 native table, change precisely the following rows. The action notation is $(\text{write}, \text{move}, \text{target})$.

| State | Read | Old action | New action |
|---|---|---|---|
| 38 | 0 | $(0, R, 28)$ | $(0, R, 11)$ |
| 38 | 1 | $(1, R, 20)$ | $(1, R, 122)$ |
| 164 | 1 | $(1, R, 98)$ | $(1, R, 130)$ |
| 214 | 0 | $(0, R, 188)$ | $(0, R, 312)$ |
| 214 | 1 | $(1, R, 312)$ | $(1, R, 20)$ |
| 312 | 0 | $(0, R, 11)$ | $(0, R, 188)$ |

All native working controls remain present. These are semantic controller changes, not six arbitrary search mutations. At source level they replace

$$\texttt{node163} = (\texttt{node6}, \texttt{node55}) \qquad \text{by } (\texttt{node6}, \texttt{node147}),$$
$$\texttt{node170} = (\texttt{node36}, \texttt{node169}) \qquad \text{by } ((\texttt{node36}, (\texttt{node167}, \texttt{node145})), \texttt{node57}).$$

The changed MP-success tail transfers its result into the cached formula; the changed Shift tests that cache and then performs the stack transfer. Section 5 states their exact numerical semantics.

The new source has 3,649 sites. Its first 3,634 instructions agree with the old source after relocating the three outgoing references at sites 48, 3,553, and 3,633. Its final fifteen instructions implement the changed controller. The builder checks every instruction and every ordinal target, including both decrement branches. More strongly, the entire array is equal to the frozen source array extracted from the current PA195C packet. Both the main builder and the independent checker assert that equality.

### 3.2 The unchanged decoder quotient

The fixed source macros and runtime use 134 controls. The selected binary association of the 190-token AX word uses 67 raw reader controls, so the unfolded machine has 201 controls. The seven disjoint identifications are

$$2016 \mapsto 2052, \quad 2018 \mapsto 2027, \quad 2022 \mapsto 2032, \quad 2024 \mapsto 2046,$$
$$2037 \mapsto 2015, \quad 2055 \mapsto 2019, \quad 2060 \mapsto 2014.$$

They are the inherited PA194-67 identifications. For each pair, incoming rows establish a different previous tape bit for its two members. The shared state can inspect that bit and use an existing helper to recover the appropriate action. Every incoming target is projected through the same seven identifications. The final count is

$$134 + 67 - 7 = 194.$$

No extra startup control, hidden marker symbol, stationary move, or nonblank initial input is introduced.

The checker exhausts the four previous-bit/current-bit combinations per pair against the *final simultaneously folded table*. This gives 28 exact local cases. It checks the complete two-cell tape, displacement, projected destination, and that no unspecified exterior cell is read. It also checks all incoming rows establishing the context. Thus changing the controller cannot invalidate a helper merely because the old pair was once individually valid.

**Lemma 3.1** (Positive quotient simulation)**.** *The raw 201-control machine and the final 194-control native machine have equivalent halting behavior from native state 74 on every coded initial tape. Each raw step has a positive finite simulation in the folded machine.*

*Proof.* At ordinary controls, the two transitions commute with the projection. At an exceptional control, its incoming row supplies the previous bit, and the corresponding checked two-cell trace supplies the simulation. The start is not exceptional. Induct on the raw trace length, preserving the complete tape and head at the endpoints. Positivity prevents an infinite raw computation from being compressed into finitely many folded steps. More concretely, if the folded table first halted after $T$ steps while the raw table had $T+1$ nonhalting steps, their positive simulations and determinism would contradict that first halt. All cases and this finite induction are available in PA. □

# 4 The exact numerical calculus retained by the construction

The physical optimization preserves a particular classical numerical calculus, rather than replacing PA by an arbitrary small checker. Its codes share the natural numbers with variable labels. Let

$$p(a,b) = \frac{(a+b)(a+b+1)}{2} + b, \qquad I(A,B) = 4p(A,B),$$

and define membership, universal quantification, and equality by

$$M(a,b) = 4p(a,b)+1, \qquad U(a,B) = 4p(B,a)+2, \qquad E(a,b) = 4p(a,b)+3.$$

In formulas below, an arrow denotes the constructor $I$. Code zero is truth, and code one is the false code $M(0,0)$. Put $N(B) = I(B,1)$, $B \wedge C = N(I(B,N(C)))$, and $\exists x_a B = N(U(a,N(B)))$. The quantifier's arguments are transposed in its numerical encoding. The exception $I(0,0)=0$ is handled explicitly: zero is decoded as truth, and the proof translations use the fixed equivalence between truth and truth implying truth. No argument assumes that all constructor images are syntactically disjoint at zero.

## 4.1 Axiom heads and gate

The six logical heads, for every $A, D, H \in \mathbb{N}$ and in the notation used by the actual emitter, are

$$\begin{aligned}
L_1 &= (\exists x_D\, U(D,H)) \to H,\\
L_2 &= U(D,U(A,H)) \to U(A,U(D,H)),\\
L_3 &= M(H,D) \to (E(H,A) \to M(A,D)),\\
L_4 &= E(H,D) \to (M(A,H) \to M(A,D)),\\
L_5 &= \exists x_D E(H,D),\\
L_6 &= U(D,A) \to \big(U(D,A \to H) \to U(D,H)\big).
\end{aligned}$$

The implicational head ranges over $D, A, H, d \in \mathbb{N}$ with $d \le D$:

$$\rho(D,A,H,d) = ((H \to d) \to D) \to ((D \to H) \to (A \to H)), \qquad d \le D.$$

The separate diagonal head, for every $B \in \mathbb{N}$, is

$$\Gamma(B) = B \to U(B,B).$$

For the guarded head, for every $a, b, c \in \mathbb{N}$, define

$$\begin{aligned}
V(a,b,c) &= M(a,b) \to U(c, M(a,b)),\\
P(a,b,c) &= c \to M(c,b),\\
S(a,b,c) &= U\big(b, (M(a,b) \to P(a,b,c)) \wedge (M(c,b) \to c)\big),\\
Q(a,b,c) &= U(c, S(a,b,c) \to N(P(a,b,c))),\\
G(a,b,c) &= N(V(a,b,c) \to Q(a,b,c)),\\
K(a,b,c) &= \begin{cases} G(a,b,c), & c \neq a \text{ and } c \neq b,\\ 0, & \text{otherwise.} \end{cases}
\end{aligned}$$

The calculus has these axiom instances, modus ponens, and generalization of theorems. The emitter additionally pushes a generalization of its old cached theorem, as specified in the next section; that entry is not a new axiom schema. The supplied development of the restricted implicational basis recovers the required classical propositional interfaces. Its proof-producing translations are part of the inherited calibration.

### 4.2 Freshness and the separation of the diagonal head

Every variable label occurring in a nonzero formula code $B$ is strictly less than $B$. Induction on decoding proves this from the strict component bounds of nonzero constructors; truth has no variables. Consequently $x_B$ is absent from the formula with code $B$, and $\Gamma(B)$ is an unconditional instance of vacuous generalization. It is safe without the two comparisons guarding $K$.

The earlier bundled-diagonal gate was

$$F_\Gamma(a,b,c) = N(\Gamma(c) \to (V(a,b,c) \to Q(a,b,c))).$$

These translations use the recovered classical calculus, including its derived ex-falso principle. To see the latter explicitly, put $E_B = \exists x_0 U(0,B) = N(U(0, N(U(0,B))))$. The recovered propositional K law gives $1 \to E_B$, while $L_1$ at $D = 0, H = B$ gives $E_B \to B$; implication composition therefore gives $1 \to B$. With this principle, the classical templates derive both $\Gamma(c)$ and $G(a,b,c)$ from the bundled formula and conversely. Every desired diagonal instance can be obtained from a guarded bundled instance with $a = b = c+1$. Thus the old bundled and current separate calculi have primitive-recursive proof translations preserving conclusions. At command level the separation adds a saved stack entry, which is why the correct ten-entry selectors must be proved instead of copied from the old nine-entry interface.

The guards on $K$ cannot be removed merely because a separation subformula looks easier when labels coincide. The same gate asserts $V(a,b,c)$. For $c = a$, its antecedent can hold for a finite-set code while universal quantification over that set variable makes the consequent false; the case $c = b$ has the analogous problem in the element position. The present construction preserves both guards.

### 4.3 Logical bridge used in the theorem

The inherited logical calibration[4, 2, 3] identifies refutations of this precise calculus with PA refutations through syntactic transformations in both directions. On the PA-to-calculus side, diagonal vacuity recovers the equality and membership interfaces needed by the inherited PA-refutation compiler. On the reverse side, membership is interpreted as binary membership of a finite-set code, equality as arithmetic equality, and quantification as first-order arithmetic quantification. Finite comprehension is

supplied by PA induction, with a uniform primitive-recursive procedure producing the proof instance for each input formula.

For definiteness the binary-membership formula may be taken as

$$\mathrm{Bit}(s,z) \equiv \exists u,v\,[s=(2u+1)2^z+v \ \wedge \ v<2^z].$$

PA proves $\mathrm{Bit}(s,z)\to z<s$, hence the negation of the translation of $M(0,0)$. It also proves, for every arithmetic formula $\theta(z,\vec{u})$,

$$\forall n\,\exists s<2^n\,\forall z<n\,(\mathrm{Bit}(s,z)\leftrightarrow\theta(z,\vec{u})).$$

The proof extends the previous code by either zero or $2^n$ according to $\theta(n,\vec{u})$. Instantiating this induction-proof pattern is a syntactic operation on formula codes, not a uniform truth predicate. When the two pointed-separation labels coincide, the inner binder changes both occurrences; the reference proof treats that case separately.

The next section gives the complete changed-controller proof compiler and states how it composes with these logical translations. The detailed inherited diagonal and separation arguments are retained in `reference/logical_calibration/` and `references/`; their exact input identities are recorded. The new physical proof does not derive the logical calibration merely from observed nonhalting.

# 5 The changed command controller and its PA calibration

The new machine retains the separate-diagonal 190-token axiom emitter and changes the controller that uses its output. It combines two earlier controller transformations: Shift tests the cached head alone, and successful modus ponens puts its conclusion in that cache. The two transformations must be composed: transplanting only one of their old proof-word compilers would give incorrect stack identities.

There is a particularly strong source-level cross-check. After resolving every structured jump, the complete 3,649-instruction program is identical to the supplied PA195C program. The frozen prior instruction array has SHA-256

`0afee1346c0931b3b91fea70571b01fd665a802ab9327409e4030f673412f5e3`.

The new physical allocation therefore implements an already calibrated source program. The proof below also describes that calibration, so that the change from the existing 3,673-instruction PA194 is explicit.

## 5.1 Stores, numerical aliases, and the two changed rules

Let

$$p(a,b)=\frac{(a+b)(a+b+1)}{2}+b, \qquad [a;w]=p(a,w), \qquad I(a,b)=4p(a,b).$$

Use iterated cons notation $[a_1,\dots,a_k;w]$, with $a_1$ on top. The total projections satisfy $\mathrm{hd}[a;w]=a$, $\mathrm{tl}[a;w]=w$, and $\mathrm{hd}(0)=\mathrm{tl}(0)=0$. In particular, $[0;0]=0$: this encoding does not distinguish an empty stack from a stack containing only zero entries. All subsequent identities are identities of numbers with these total projections.

A command store is $\langle t, w, d, h\rangle$, consisting of top formula, underlying stack, numerical offset, and cached formula. The modified Shift is

$$S\langle t, w, d, h\rangle = \begin{cases} \mathsf{halt}, & h = 1, \\ \langle h, \mathrm{tl}(w), d, \mathrm{hd}(w)\rangle, & h \neq 1. \end{cases} \tag{C1}$$

For $y = d + x$, the modified modus-ponens command is

$$\mathsf{MP}_x\langle t, w, d, h\rangle = \begin{cases} \langle 0, w, 0, y\rangle, & I(t, y) = h, \\ \langle 0, w, y, 0\rangle, & I(t, y) \neq h. \end{cases} \tag{C2}$$

Only Shift can halt. The original PA194 controller tests $t + h = 1$ at Shift and returns $\langle y, [0; w], y, 0\rangle$ on successful MP. Thus the new controller has different command semantics, rather than only renamed physical controls. For example, $I(0, 2) = 20$, and MP with operand two sends $\langle 0, 0, 0, 20\rangle$ to $\langle 0, 0, 0, 2\rangle$ in the new controller and to $\langle 2, 0, 2, 0\rangle$ in the original one.

Write $U(a, B) = 4p(B, a) + 2$ and $\Gamma(B) = I(B, U(B, B))$. For $D = d + x$, $H = h + y$, and $A = c$, the unchanged AX command is

$$\mathsf{AX}_{c,x,y}\langle t, w, d, h\rangle = \langle K(H, D, c), [q_1, \ldots, q_9, t; w], 0, 0\rangle, \tag{C3}$$

where, from stack top to bottom,

$$(q_1, \ldots, q_9) = (\Gamma(A), L_6, L_5, L_4, L_3, L_2, L_1, \rho(D, A, H, d), U(d, h)).$$

Here $L_1, \ldots, L_6$ are exactly the six inherited logical heads, and

$$\rho(D, A, H, d) = ((H \to d) \to D) \to ((D \to H) \to (A \to H))$$

denotes the corresponding numerical implication code, with $d \leq D$. The top $K$ is the inherited separation/vacuity gate, replaced by zero when its numerical freshness guard fails. The guard does not suppress the separately pushed diagonal instance. Every $q_j$ except $q_9$, and also $K$, has residue zero modulo four; $q_9$ has residue two. Consequently none of these generated entries is the false code one. This is an arithmetic fact about constructors and requires no consistency assumption. The complete 190-token emitter word, not just its head set, is unchanged from PA194.

### 5.2 Uniform stack identities for ten pushed entries

Commands in a word execute from left to right. Iterating (C1) after (C3) gives, for every incoming store,

$$\begin{aligned} \mathsf{AX}_{c,x,y}; S^9 : \langle t, w, d, h\rangle &\longmapsto \langle \rho(d + x, c, h + y, d), [t; w], 0, U(d, h)\rangle, \\ \mathsf{AX}_{c,x,y}; S^{10} : \langle t, w, d, h\rangle &\longmapsto \langle U(d, h), w, 0, t\rangle. \end{aligned}$$

The first Shift tests zero; the next nine test generated entries. None of these ten tests examines the saved top $t$. These identities therefore hold even when $t = 1$ and at every zero-stack alias.

Define

$$\begin{aligned} \mathsf{Push} &= \mathsf{AX}_{0,0,0}; S^9; \mathsf{MP}_0, \\ \mathsf{Pop} &= \mathsf{AX}_{0,0,0}; S^{11}, \\ \mathsf{Discard} &= \mathsf{MP}_1; \mathsf{AX}_{1,0,0}; S^{10}; \mathsf{MP}_0. \end{aligned}$$

They satisfy

$$\begin{aligned}\mathsf{Push}\langle t,w,d,h\rangle &= \langle 0,[t;w],0,0\rangle,\\ \mathsf{Pop}\langle t,[a;w],d,h\rangle &= \begin{cases}\mathsf{halt}, & t=1,\\ \langle t,w,0,a\rangle, & t\neq 1,\end{cases}\\ \mathsf{Discard}\langle t,w,0,0\rangle &= \langle 0,w,0,0\rangle.\end{aligned}$$

For Push, its last comparison is between an implication code and the universal code $U(d,h)$, so MP fails with offset zero. Pop's last Shift tests the saved top and, if safe, removes the required underlying entry. For Discard, $I(t,1)>0$ forces its first MP to fail; the AX identity then gives $\langle U(1,0),w,0,0\rangle$, and $I(U(1,0),0)>0$ forces its final MP to fail with offset zero. Pop's possible early halt is intentional and is included in the proof compiler, rather than excluded by an unproved noncontradiction invariant.

Let $\mathsf{Offset}_0$ be empty and $\mathsf{Offset}_i=\mathsf{MP}_i$ for $i>0$. It maps $\langle 0,w,0,0\rangle$ to $\langle 0,w,i,0\rangle$. The word

$$\mathsf{Gen}_i=\mathsf{Push};\mathsf{Offset}_i;S;\mathsf{AX}_{i,0,0};S^{10}$$

maps $\langle a,w,0,0\rangle$ to $\langle U(i,a),w,0,0\rangle$ for every $a,w$, including $a=1$. The Shift before AX tests cache zero and fetches $a$ into the cache; AX clears that cache before it could be tested.

For $1\le j\le 7$, the selector

$$\mathsf{Discard};\mathsf{AX}_{A,D,H};S^{j+1};\mathsf{Pop}^{9-j}$$

returns $\langle q_j,w,0,0\rangle$ from $\langle t,w,0,0\rangle$. This includes the diagonal head at $j=1$. Each Pop preserves a generated implication code and hence is safe. For $u\le a$, the restricted propositional selector is

$$\mathsf{Discard};\mathsf{Offset}_u;\mathsf{AX}_{b,a-u,c};S^9;\mathsf{Pop},$$

and the gate selector is $\mathsf{Discard};\mathsf{AX}_{c,b,a};\mathsf{Pop}^{10}$. These return respectively $\rho(a,b,c,u)$ and $K(a,b,c)$, preserving the arbitrary underlying stack and clearing both caches.

The new MP suffix is particularly short:

$$\mathsf{MPSuffix}_b=S;S;\mathsf{MP}_b;S.$$

From $\langle 0,[a,I(a,b),0;w],0,0\rangle$, it halts if $a=1$ or $b=1$, and otherwise returns $\langle b,w,0,0\rangle$. The first Shift fetches $a$; the second tests it and fetches the major premise; MP puts $b$ in the cache; the final Shift tests $b$ and consumes the reserved zero. Without that zero, the suffix would consume an entry of the framed stack. Total Cantor decoding proves the same identity when $a=b=w=0$.

### 5.3 Completeness and soundness at the command level

Let $\mathcal{C}$ be the numerical separate-diagonal calculus implemented by the unchanged emitter, with its inherited restricted propositional axioms, generalization of theorems, and modus ponens. For each finite proof tree $p$ of $\mathcal{C}$, construct a word $W_p$ which, from every $\langle t,w,0,0\rangle$, either halts or returns $\langle \mathrm{concl}(p),w,0,0\rangle$. Use the displayed selectors at axiom leaves and $W_{p_0};\mathsf{Gen}_i$ at a generalization node. At an MP node with minor premise $p_0$ proving $a$ and major premise $p_1$ proving $I(a,b)$, use

$$\mathsf{Discard};\mathsf{Push};W_{p_1};\mathsf{Push};W_{p_0};\mathsf{Push};\mathsf{MPSuffix}_b.$$

The first two words reserve the zero required by the MP suffix. Induction on the proof tree proves the stated contract. Every early halt is allowed, and any suffix appended to an already halting word preserves that halt. If $p$ proves one, then

$$\mathsf{AX}_{0,0,0};W_p;\mathsf{Pop}$$

halts from every incoming store, including stores with arbitrary caches: the prefix AX clears both caches, and the final Pop tests the conclusion if an earlier command has not already halted. The inherited scheduler enumerates every finite command word. The universal incoming-store property is essential because previous trials can leave a store behind.

For the converse, annotate the top $t$, cached formula $h$, and every live canonical stack entry with a $\mathcal{C}$ proof. No derivability condition is imposed on the numerical offset $d$. For $w > 0$ one has $\mathrm{tl}(w) < w$, so repeated tail projection reaches zero in at most $w$ steps and gives a primitive-recursive finite canonical stack. The fixed proof $\rho(0,0,0,0) = 0$ supplies the implicit zero tail. Thus $[0;0] = 0$ requires no infinite annotation.

AX saves the annotated top, generalizes the old cached formula, and emits axioms; its dominance condition is $d \leq d + x$. MP failure returns the proved code zero, and MP success applies modus ponens to its two annotated premises. Shift moves annotated entries, and a halting Shift exposes an annotated one. Induction on the finite command trace proves that every source halt yields a $\mathcal{C}$ refutation. The new command contracts, the selectors, the proof-tree compiler, and all annotation updates use primitive-recursive operations. PA verifies their defining equations and the indicated finite inductions.

### 5.4 The inherited logical bridge to PA

The logical calibration is inherited from the separate-diagonal PA194 and PA195C analyses[4, 2, 3]; its full argument is retained with the reference material. Its two directions are syntactic proof transformations, not an appeal to external soundness of PA. Here are the interfaces used by the present construction.

For completeness, the diagonal schema $\Gamma(B) = B \to \forall x_B B$ replaces equality vacuity. Every variable label in a formula of code $B$ is smaller than $B$. To recover reverse membership rewriting, put $C = E(a,b) \to (\chi(b) \to \chi(a))$, take the separator property $\neg C$, and choose its own numerical code as the fresh separator label. Pointed separation, the retained direct membership rewrites, the equality witness, and membership-only vacuity imply the reverse rewrite; the instance $\Gamma(\neg C)$ discharges the separator without presupposing equality vacuity. A diagonal instance for $E(a,b)$ gives the neighborhood equality interface at a fresh binder, and the witness and two-way membership rewrites transport that binder. These are the interfaces needed by the inherited PA-refutation compiler. All choices of fresh labels, formula constructions, case splits for coincident labels, and finite proof concatenations are primitive recursive and PA-verifiable.

For soundness, interpret membership by binary membership of a finite-set code, equality by arithmetic equality, and quantification by ordinary first-order quantification. PA proves finite binary comprehension for each arithmetic formula by induction on the finite bound. Substitution of the formula syntax into that fixed induction-proof template gives a primitive-recursive proof-producing operation whose correctness PA verifies; this is not a uniform arithmetic truth predicate. Pointed separation is an instance of finite comprehension, and diagonal vacuity uses the strict code bounds. The label-collision case $a = b$ is handled separately, since the inner binder changes both occurrences of that label. Code one translates to binary self-membership, which PA refutes. The constructor alias $I(0,0) = 0$ is handled by the fixed proof equivalence of truth and truth implying truth.

Consequently the inherited transformations give

$$\mathrm{PA} \vdash (\exists p\ \mathrm{Prf}_{\mathcal{C}}(p,1)) \iff (\exists q\ \mathrm{Prf}_{\mathrm{PA}}(q, \ulcorner 0 = 1 \urcorner)).$$

Combining this with the command-level constructions gives the corresponding equivalence for halting of the exact 3,649-instruction source program. The physical verification then supplies every source-

instruction simulation for arbitrary natural register values and arbitrary finite nonempty spare lists, together with the exact blank-tape startup. Positive simulation durations allow finite traces to be composed in both directions. All fixed local cases and the generic list-sweep lemmas are formalizable in PA. Therefore, for the delivered literal table,

$$\boxed{\mathrm{PA} \vdash \neg\,\mathrm{Halt}(\mathrm{PA194C}, \mathrm{blank}) \iff \mathrm{Con}(\mathrm{PA}).}$$

This is an ordinary mathematical proof of a PA-provability statement, supported by reproducible finite and symbolic certificates. The package does not claim a newly elaborated end-to-end Lean or Rocq theorem.

### 5.5 What transfers from ZFC and concise iPRA research

The useful ZFC ideas are the substitution-free numerical calculus, small register-machine architecture, careful separation of source correctness from literal tape simulation, and explicit condensed-detachment certificates for the propositional basis. The inspected `lukasiewicz_basis.md` records K, S, Peirce, composition and ex-falso interfaces, while also distinguishing a named-formula completeness theorem from its numerical and machine bridges. The same distinction governs the present proof obligations.

The concise-iPRA work illustrates how a small family of proof rules can recover larger logical interfaces by derivation, rather than by encoding each as an independent primitive. That is relevant to future emitter redesign, but its theorem is calibration to iPRA. A compact iPRA proof checker does not by itself establish the required translations to and from PA refutations, nor a binary-machine state bound. The present result therefore retains the PA-calibrated emitter and obtains a different 194-state construction by combining its allocation with the changed controller.

## 6 Physical correctness and its verification boundary

The source uses nine natural-number registers. The physical file represents a value $v$ by $0\,1^{v+1}$, followed by an arbitrary finite nonempty spare list. Its complete source frame has a left sentinel, a binary instruction path, a zero gap, a right delimiter, and this register file. The base header length is 48; the gap may grow during computation. The symbolic contracts quantify over the gap and register values instead of choosing a finite collection of sample numbers.

For an increment site, the certificate reaches the next source frame with precisely the selected register incremented. For a decrement site, it separately checks the zero and positive cases, including the right successor and the changed register value. Jumps preserve the register file and change the code path; the halt primitive reaches native label 254. The complete instruction graph fixes every such destination.

The first checker uses the inherited generic arithmetic-block lemmas and verifies all 4,749 outcomes for both the raw 201-control and final 194-control tables. The second checker uses a separate literal-tape representation and derives four generic list-sweep lemmas from the actual transition rows. It then checks every final-table source outcome without calling the first checker's arithmetic-box oracle. It independently resolves the source graph, verifies the dense/native conjugacy, and checks the entire finite-support blank-start tape.

| Verified quantity | Value |
|---|---|
| Working states / transition rows | 194 / 388 |
| Raw controls / compatible folds | 201 / 7 |
| Exact local quotient cases | 28 |
| Source primitive sites | 3,649 |
| Universal source outcomes in each physical pass | 4,749 |
| Independent symbolic steps | 1,096,302 |
| Independent maximum symbolic steps per outcome | 403 |
| Generic list lemmas rederived | 4 |
| Literal transition rows exercised by the independent audit | 388 |
| Exact blank-start initialization | 57,922 transitions |
| Symbolic command-algebra cases | 188 |

The symbolic step totals count proof-execution operations, including generic sweep steps; they are not uniform literal running-time bounds for arbitrary unary register values. The finite local fold traces, by contrast, are literal transition sequences on their stated finite window.

At the startup endpoint the native state is 84, the dense state is 24, and the head is at $-2$. The first nine physical values are $(0, 0, 0, 0, 0, 0, 2, 0, 0)$ and the spare list has 26 entries equal to one. There are 66 one-cells in the entire tape; every other cell is zero. The machine-readable startup certificate lists all 66 coordinates. Equality is checked for the whole finite-support tape, not only a displayed window.

**Proposition 6.1** (Source/physical halting equivalence)**.** *PA proves that the delivered blank-tape machine halts if and only if the exact Combined source program halts from its certified clean seed.*

*Proof.* Execute the fixed startup trace. Compose the positive primitive simulations using induction on the source trace. The frame invariant preserves the representation of the evolving register store, the complete tape, and the nonempty spare list. For the converse, suppose the physical table first halts at time $T$. If the source has not halted, its total primitive step function generates $T + 1$ further source steps. Their positive simulations give a nonhalting physical prefix longer than $T$, contradicting determinism. Fixed local cases are finite arithmetic facts; unary and finite-list sweeps are proved by induction. Hence the argument concerns coded finite traces inside PA. Compose it with the source calibration in Section 5 to obtain the main theorem. □

The two executable checkers are supporting proof artifacts, not two independent foundational proof kernels. They use related source encodings and ordinary Python. The second removes the first one's arithmetic-oracle dependency and checks the generic sweeps directly, which addresses a specific possible blind spot. No claim of a new fully elaborated Lean or Rocq endpoint is made.

## 7 Material difference and exact graph comparison

The controller changes what commands do: successful MP stores the conclusion in a different component, and Shift has a different halt test. The expanded source loses 24 instructions. Thus the distinction is observable before physical state allocation. The numerical example in Section 5 exhibits the different store results.

The physical distinction is separately certified. The corpus contains seven exact 194-state byte forms, representing four prior action-labeled graphs: three positive designs and the rejected scanner quotient. For each table the graph checker starts a deterministic breadth-first numbering from every working state, following the read-zero edge before the read-one edge. Every one of the 194 possible roots spans

the whole graph. It compares the resulting canonical transition vectors directly, taking the least over all roots; equality therefore characterizes graph isomorphism for these tables without relying on the given start state.

The same calculation is repeated under left/right reflection and global bit exchange. PA194C differs from every previous graph under all these symmetries. Bit exchange is only an extra structural comparison—the actual construction retains blank symbol zero. The comparison does not claim that graph nonisomorphism by itself proves a different algorithm; the changed command semantics and full source graph provide that evidence.

# 8 The search below 194 states

The search objective was a complete physical machine with at most 193 working states. An optimistic reader count, a local fold in isolation, or a long nonhalting blank prefix was not an acceptance criterion. No below-194 result survived the required construction and proof checks.

## 8.1 Association, history folds, and the 194-state plateau

The two existing allocations suggest the numerical target of 59 effective AX readers: 66 raw readers with seven folds, 67 with eight, or 68 with nine, while retaining 134 fixed controls. We started independent searches from both PA194 trees, preserving the 190-token emitter word. Candidate scores used compatible history folds checked on the simultaneous quotient, with scanner legality included. Further runs substituted the Combined fixed rows, allowed mixed fixed-reader/AX-reader pairs, varied actual head-parameter and ordering choices, and applied an additional unconditional exact-row quotient after the contextual quotient.

Two finite traversals closed their recorded score-194 queues under the stated single-occurrence and shared-subtree rotation moves: one used the original controller and ordinary folds, the other Combined and mixed folds. Each processed 11,472 parent grammars, evaluated 2,373,136 new neighbors, and retained 11,470 nonseed successors. Neither reached the 20,000-entry queue cap. Both found neither a score below 194 nor a raw-reader count below 67. Deduplication uses a 64-bit association hash; queue closure is therefore a recorded computation, not a collision-free mathematical exhaustiveness certificate. In particular, it is not a global grammar or machine lower bound.

The eight annealing runs contain 49,284,000 completed logged proposal slots, with repeated proposals and repeated machines included. Only complete logged progress intervals are counted; interrupted unlogged tails are excluded. Their recorded best zero-penalty state count is 194.

| Search | Logged proposal slots |
|---|---:|
| Original controller, baseline A | 10,383,000 |
| Original controller, baseline B | 9,123,000 |
| Combined, ordinary AX folds | 4,143,000 |
| Actual head parameters and head order | 8,043,000 |
| Original controller, mixed folds | 5,163,000 |
| Combined, mixed folds | 5,403,000 |
| Combined plus zero-jump, mixed folds | 4,503,000 |
| Combined plus zero-jump, extra exact-row closure | 2,523,000 |

The broader zero-jump/extra-closure plateau traversal was stopped after 5,600 logged parent expansions and 1,163,952 new neighbors, with 1,688 grammars still pending; that traversal was not exhausted.

Complete code, inputs, seeds, finite replay arguments, logs, and input bindings are under `research/association/`. Proposal slots are not counts of fully verified machines. No lower bound is inferred from any of these negative results.

### 8.2 Direct runtime searches

We also searched the two supplied principal 194 tables and PA194C directly. The local-action procedure considers replacing a transition by another first action followed by a bounded detour through existing controls. It then asks whether two non-box controls can share both replacement actions without one detour relying on the other removed control. Every claimed local match must restore the entire inspected tape window and the required head displacement and destination.

Both conservative all-tape contexts and stronger contexts extracted from all symbolic source outcomes plus exact blank startup were tried. The radius-four runs use a nine-cell window and a 100-transition detour bound. The PA194C radius-six run uses a thirteen-cell window and a 160-transition bound. The extracted pattern counts were 8,338 for PA194-67, 8,873 for PA194-68, 8,240 for PA194C at radius four, and 29,107 for PA194C at radius six. No additional pair was emitted in these runs. The arithmetic-box controls were excluded as merged controls so that their established contracts remained applicable.

A separate raw-table scan enlarged the pair domain to fixed and AX readers, using *all* incoming rows to establish the previous bit. At the two baseline grammars, this revealed no mixed fixed/AX fold beyond the known AX-only opportunities. Later association searches applied the larger domain to changing trees. Failure within these templates does not rule out a different encoding, a larger simultaneous rewrite, or a stronger reachable-configuration invariant.

### 8.3 A specific rejected scanner shortcut

Removing native scanner control 34 and redirecting its incoming transitions to 75 would give 193 working controls. It shortens the header test by one zero. In PA194C, however, 113 of the 4,749 arbitrary-register source outcomes fail the existing generic simulation contract. For example, source site zero has address `000000`, and the clean entry site 25 has address `1000000`; the shortened scanner cannot be justified by the old address invariant. The complete failed-site list is preserved.

This rejects the shortcut under the proposed proof method. The failures are not advertised as a proof that every failing frame is reachable from blank or that the literal shortcut cannot have some other consistency interpretation. A separate two-million-transition blank screen did not halt, which supplies no missing correctness argument. The machine is therefore not accepted as the requested smaller construction.

### 8.4 Additional 194-state seed and remaining directions

The supplementary PA194CZ table composes PA194C with the previously studied zero-register jump at source site 26. It changes native row $(27, 1)$ from $(0, L, 40)$ to $(1, R, 21)$ and replaces that source jump by a decrement of register six. A common inductive set of 263,244 site/positivity-mask pairs proves that register zero test always takes the required branch there. It has 4,750 physical branch contracts and a 58,168-step blank startup. This is another 194-state search seed, not the principal result or an additional state saving. It has the primary construction/invariant checks, not a separately completed new independent physical replay.

The useful next changes would affect the joint source/decoder objective, or a fixed-runtime invariant strong enough to share an additional control. Deleting a logical head requires both soundness and a replacement refutation compiler. Deleting a freshness comparison is particularly unsafe because the guarded gate also contains membership vacuity, as explained in Section 4. None of the negative searches in this report establishes mathematical optimality of the current count.

# 9 Reproduction and delivered artifacts

Run the main acceptance checks from the unpacked archive root:

```
python3 -B reproduce.py
```

The wrapper runs deterministic construction and source binding, both symbolic physical passes, the independent final-table audit, the symbolic command-algebra audit, and the complete existing-194 novelty check. Python 3 and its standard library are sufficient. Run with assertions enabled; optimization mode is rejected. The main checks do not need the repository checkout, internet access, a solver, or a proof assistant.

The optional discovery searches are separate. Their C++ sources and frozen inputs are included with the run records, but replaying heuristic search is not required to reconstruct or verify the delivered machine. The complete corpus census requires the pinned repository because the large incoming corpus is identified by hashes rather than duplicated.

| Location | Purpose |
|---|---|
| `construction/tables/` | Definitive dense/native tables and exact label map. |
| `construction/source/` | Grammar, seven folds, all primitive sites and instructions. |
| `construction/evidence/` | Complete primary verification and startup receipts. |
| `independent/` | Separate literal-tape checker, derived lemmas, branch traces. |
| `comparison/` | All-archive census, family ledger, source identity, graph novelty. |
| `references/` | Preserved logical and command-calibration dependencies. |
| `research/` | Below-194 search inputs, code, records, and rejected shortcut evidence. |
| `report/` | This article, all its LaTeX inputs, and the compiled PDF. |

The inherited calculus, register engine, compiler, and prior optimizations retain their attribution and license notices. The contribution here is their verified combination at 194 states, its explicit table, fresh universal physical checks, exact comparison, and documented search.

# A Complete transition table

The following table is the complete `pa194c.tm`. An action $wDn$ writes $w$, moves in direction $D$, and enters state $n$. Start in state 0 on the all-zero tape; 194 is the halt label. Two working states are printed per row only to save space.

| State | Read 0 | Read 1 | State | Read 0 | Read 1 |
|---|---|---|---|---|---|
| 0 | 1R1 | 1R2 | 97 | 0R36 | 1R112 |
| 1 | 1R3 | 1R1 | 98 | 0R111 | 1R75 |
| 2 | 0R0 | 1R2 | 99 | 0L63 | 0R109 |
| 3 | 0R4 | 0R1 | 100 | 0L62 | 1R113 |
| 4 | 1R5 | 1L6 | 101 | 0L90 | 1R114 |
| 5 | 1L7 | 1L8 | 102 | 0R115 | 0L14 |
| 6 | 0L7 | 1L6 | 103 | 0R69 | 1R103 |
| 7 | 0L9 | 1L6 | 104 | 0R116 | 1R95 |
| 8 | 0L10 | 0L11 | 105 | 0R116 | 1R67 |
| 9 | 1R12 | 0L9 | 106 | 0R117 | 0L14 |
| 10 | 0L9 | 0L10 | 107 | 0R118 | 1L71 |
| 11 | 1L8 | 1L11 | 108 | 0R119 | 1R120 |
| 12 | 0R12 | 0R13 | 109 | 0R14 | 1R109 |
| 13 | 0L14 | 0R13 | 110 | 0R66 | 1R110 |
| 14 | 1L15 | 0L15 | 111 | 0R121 | 1L35 |
| 15 | 0L15 | 1L16 | 112 | 0R122 | 1R123 |
| 16 | 0L17 | 1L16 | 113 | 0R88 | 0L14 |
| 17 | 0L18 | 1L16 | 114 | 0R90 | 1L95 |
| 18 | 0L19 | 1L16 | 115 | 0R124 | 1R124 |
| 19 | 0L20 | 1L16 | 116 | 0L63 | 0R103 |
| 20 | 0L21 | 1L16 | 117 | 0R68 | 0R91 |
| 21 | 0L22 | 1L16 | 118 | 0R58 | 0R91 |
| 22 | 0L23 | 1L16 | 119 | 0L86 | 1R125 |
| 23 | 0R23 | 1R24 | 120 | 0R126 | 1R127 |
| 24 | 0R25 | 1R26 | 121 | 0R58 | 0R110 |
| 25 | 0R27 | 1R28 | 122 | 0R99 | 1R113 |
| 26 | 0R29 | 1R30 | 123 | 0R79 | 1R78 |
| 27 | 0R31 | 1R194 | 124 | 0R88 | 1R88 |
| 28 | 0L32 | 1R33 | 125 | 0R128 | 1R129 |
| 29 | 0R34 | 1R35 | 126 | 0R32 | 1R36 |
| 30 | 0L36 | 1L37 | 127 | 0R31 | 1R111 |
| 31 | 0R38 | 1R39 | 128 | 0R130 | 1R131 |
| 32 | 0R40 | 1R41 | 129 | 0R132 | 1R133 |
| 33 | 0R42 | 1R43 | 130 | 0R134 | 1R135 |
| 34 | 0R44 | 1R45 | 131 | 0R136 | 1R137 |
| 35 | 0R46 | 1R33 | 132 | 0R138 | 1R139 |
| 36 | 0R47 | 1R48 | 133 | 0R140 | 1R141 |
| 37 | 0R49 | 1R50 | 134 | 0R97 | 1R35 |
| 38 | 0R51 | 1R42 | 135 | 0R142 | 1R124 |
| 39 | 0R52 | 1R53 | 136 | 0R143 | 1R144 |
| 40 | 0R54 | 1R55 | 137 | 0R145 | 1R146 |
| 41 | 0R56 | 1R57 | 138 | 0R134 | 1R147 |
| 42 | 0R58 | 0R59 | 139 | 0R88 | 1R148 |
| 43 | 0R60 | 0L14 | 140 | 0R149 | 1L131 |
| 44 | 0R61 | 1R62 | 141 | 0R150 | 1R151 |
| 45 | 0R63 | 1R52 | 142 | 0R76 | 1R41 |
| 46 | 0R64 | 1R65 | 143 | 0R28 | 1R152 |

| State | Read 0 | Read 1 | State | Read 0 | Read 1 |
|---|---|---|---|---|---|
| 47 | 0R58 | 0R66 | 144 | 0L153 | 1L154 |
| 48 | 0R67 | 0L14 | 145 | 0L129 | 1R155 |
| 49 | 0R68 | 0R69 | 146 | 0R156 | 1R157 |
| 50 | 0R70 | 0L48 | 147 | 0R149 | 1R158 |
| 51 | 0R64 | 1R53 | 148 | 0R159 | 1R88 |
| 52 | 0L71 | 0L15 | 149 | 0R97 | 1R160 |
| 53 | 0R72 | 0L10 | 150 | 0R161 | 1R162 |
| 54 | 0R73 | 1R74 | 151 | 0R163 | 0L10 |
| 55 | 0R75 | 1R76 | 152 | 0R164 | 1R165 |
| 56 | 0R77 | 1R78 | 153 | 0R166 | 1R157 |
| 57 | 0R79 | 1R80 | 154 | 0R167 | 1R156 |
| 58 | 0L81 | 0R1 | 155 | 0R135 | 1R130 |
| 59 | 0R82 | 1R59 | 156 | 0R147 | 1R41 |
| 60 | 0L73 | 1L46 | 157 | 0R134 | 1R168 |
| 61 | 0R83 | 1R84 | 158 | 0R41 | 1L169 |
| 62 | 0R54 | 1R85 | 159 | 0L170 | 1R171 |
| 63 | 0L86 | 1R87 | 160 | 0R107 | 1R85 |
| 64 | 0R58 | 0R82 | 161 | 0R172 | 1R170 |
| 65 | 0R88 | 1R30 | 162 | 0R173 | 1R174 |
| 66 | 0R89 | 1R66 | 163 | 0R175 | 1R176 |
| 67 | 0L90 | 1L33 | 164 | 0R177 | 1R178 |
| 68 | 0L81 | 0R2 | 165 | 0L179 | 1L180 |
| 69 | 0R91 | 1R69 | 166 | 0R149 | 1R140 |
| 70 | 0R72 | 1R92 | 167 | 0R165 | 1R181 |
| 71 | 0R46 | 1R93 | 168 | 0L130 | 1R88 |
| 72 | 0R42 | 0L13 | 169 | 0R182 | 1R135 |
| 73 | 0R94 | 1R37 | 170 | 0R183 | 1R145 |
| 74 | 0R30 | 1R95 | 171 | 0R158 | 1R130 |
| 75 | 0R42 | 1R96 | 172 | 0R139 | 1R165 |
| 76 | 0R97 | 1R98 | 173 | 0R158 | 1R184 |
| 77 | 0R99 | 1L100 | 174 | 0R185 | 1R107 |
| 78 | 0R30 | 1R77 | 175 | 0R31 | 1R52 |
| 79 | 0R93 | 1R101 | 176 | 0R186 | 1R175 |
| 80 | 0R30 | 1R102 | 177 | 0R134 | 1R142 |
| 81 | 1R81 | 1L15 | 178 | 0R32 | 1R187 |
| 82 | 0R103 | 1R82 | 179 | 0R144 | 1R188 |
| 83 | 0R94 | 0L7 | 180 | 0R189 | 1R190 |
| 84 | 0R104 | 1R105 | 181 | 0R41 | 1R191 |
| 85 | 0R42 | 1R106 | 182 | 0R168 | 1R41 |
| 86 | 0R107 | 1R48 | 183 | 0R192 | 1R88 |
| 87 | 0R108 | 0L72 | 184 | 0R153 | 1R146 |
| 88 | 0R68 | 0R66 | 185 | 0R165 | 1R180 |
| 89 | 0R109 | 1R89 | 186 | 0R92 | 1R111 |
| 90 | 0R49 | 1R100 | 187 | 0R193 | 1R159 |
| 91 | 0R110 | 1R91 | 188 | 0R169 | 1R154 |
| 92 | 0R30 | 1R48 | 189 | 0R158 | 1R192 |
| 93 | 0R47 | 1R60 | 190 | 0R179 | 1R182 |
| 94 | 0L63 | 0R89 | 191 | 0R182 | 1R41 |
| 95 | 0R94 | 1L111 | 192 | 0R149 | 1R168 |
| 96 | 0R101 | 0L14 | 193 | 0R76 | 1R144 |

# B All 36 incoming archives

The ledger records the source-level or physical contribution and its reported/checked status. Diagnostic small tables do not become accepted consistency constructions merely by appearing in the numeric census.

| Archive | Contribution and status |
|---|---|
| `Below_U208_Research_2026-10-29.zip` | Full-frame windows, offset quotients and guarded pruning; source 3,909 → 3,904; no accepted below-208 table. |
| `ConPA_below208_research_2026-10-29.zip` | Finite-model role-collapse objection, 279,936 fixed-association parameter cases, fold synthesis; no sub-208. |
| `PA195_ConPA_research_2026-11-14.zip` | Positive 195S and 195C: unchanged separate 190-token AX word, 67-6 readers; full source/runtime/startup certificates plus independent literal-tape audits. |
| `PA_consistency_TM_research_2026-11-13.zip` | 159-node five-zero source and 157-node combined representation; exact source CFG checks, no smaller physical allocation. |
| `U208_ConPA_2026-10-29.zip` | Particular native 172/211 history fold reduces a U209 parent to U208. |
| `U208_ConPA_Conditional_Context_2026-10-29.zip` | Conditional-entry proof; same two original U208 graph classes. |
| `U208_ConPA_Entry_Context_2026-10-29.zip` | Full entry-context proof; same two original classes. |
| `U208_ConPA_History_2026-10-29.zip` | Finite-history proof with four exceptional blocks; same original saving. |
| `conpa_tm_comparison_and_reduction_search.zip` | Experimental MP-result-in-H and MP-without-zero-push 196 variants; rejected Shift layouts halt at 43,563. |
| `conpa_tm_comparison_and_search.zip` | A/D register-swap 196; rejected two-rotation 195 fails 93 contracts. |
| `conpa_tm_study_2026-11-13.zip` | Combined 196 = H-only Shift + MP-result-in-H: 3,788 sites, 4,928 contracts, 57,922 startup; direct-negation 197 too. |
| `pa194_conpa_2026-11-13(2).zip` | Independent 194 = 134+68-8 allocation of exactly the same separate190 source; 32 local cases and 4,780 raw/folded contracts replayed. Also a distinct supplementary 195 layout. |
| `pa194_conpa_2026-11-13.zip` | Primary 194 = 134+67-7, same separate190 source; additional A/D-register 194 companion. Full independent physical certificates replayed in this audit; ordinary PA proof supplied, no new complete Lean/Rocq endpoint. |
| `pa195_conpa_2026-11-13.zip` | Positive original PA195: unchanged diagonal 191-token program, 68-7 reader allocation, detailed ordinary PA calibration. |
| `pa195_diagonal_combined_2026-11-13.zip` | PA195DC combines bundled diagonal AX with Combined controller: 3,676 sites, 4,784 contracts, 57,922 startup; ordinary PA consistency argument. Two secondary 195 implementation witnesses are not primary endpoint claims. |
| `pa195_followup_research.zip` | M195S8 uses eight-zero scanner and eight folds (203-8=195), with 4,815 raw/folded contracts and 59,164 startup. Rejected194 has valid local folds but an invalid scanner address. |
| `pa196_research_2026-11-13.zip` | Exact 157-node representation of 3,812-site original program; loses contextual sharing, no physical reduction. |
| `pa208_research_2026-10-29.zip` | Eighteen further helper layouts; 27 designated U208 presentations give 20 graphs, not two. |
| `pa_below196_research.zip` | Eight equivalent gates, membership/equality orientations, broad grammar searches; original 201-token best 196. |
| `pa_consistency_comparison_2026_11_13.zip` | Modal-head/explosion split, twelve semantic compositions, joint gate; physical results 196–200. |
| `pa_consistency_comparison_and_new196_2026-11-13.zip` | Diagonal 196 (3,700 sites), Separate 196 (3,673), Scanner-best 196; principal logical/source parents of positive 195. |
| `pa_consistency_cross_archive.zip` | Four new 196-state combinations: diagonal/separate AX + Combined controller, each optionally zero-jump; sites 3,676/3,649 and startups 57,922/58,168. |

| Archive | Contribution and status |
|---|---|
| pa_consistency_reduction.zip | Rebracketed 196, H-only 196, integrated gate 203/206; two failed 195 diagnostics. |
| pa_consistency_reduction_audit_2026-11-14.zip | Four further experimental MP-H compositions, two 196-state and two 197-state; parameter/order/association/helper/scanner negative searches. |
| pa_consistency_research.zip | Two exact 158-node sources and repaired 196; nominal 195 needs added scanner state. |
| pa_consistency_research_2026-10-29.zip | Inductive 3,909 → 3,904 specialization; no physical sub-208. |
| pa_consistency_research_2026-11-13(2).zip | Twelve-rotation 196, diagonal 198, successful-MP halting-source experiment; no sub-196. |
| pa_consistency_research_2026-11-13.zip | Earlier diagonal-vacuity 199 and short-negation 197. |
| pa_consistency_research_2026-11-14.zip | NEG18: INC7; INC7; node64, 18-site encoded negation; complete source 3,441 and 197 states (134+68-5); independent macro and scratch-zero proofs. |
| pa_tm_comparison_and_reductions.zip | Two rotated 196 graphs; distributed-gate 198 with 3,773 source sites. |
| pa_tm_extended_research_2026-11-13.zip | Four closing-decrement variants and stronger zero invariants; all change the same row, hence not stackable. |
| pa_tm_research_2026-11-13(2).zip | Zerojump 196 (4,959 contracts, 58,402 startup); parity/disequality experiments; rejected 195 fails source 26. |
| pa_tm_research_2026-11-13.zip | Shorter-path 196 and independent physical checking; detailed direct PA reverse-translation strategy. |
| pa_unary_power_cover.zip | Unary weak-covering theory and PA proof translations; no new compiled small table. |
| u208_comparison_research_2026-10-29.zip | Original two-class audit; weak/exact schemas; rejected 207. |
| u208_research_2026-10-29.zip | Primary/alternative U208 and symbolic contracts; same initial history fold. |

# C Sources and precise dependency boundary

## References

[1] Vladimir Reshetnikov, *ProveIt*, pinned revision `768c2be1a8e4d2fcf8aa3443c7b0f084d145734c`. https://github.com/VladimirReshetnikov/ProveIt/tree/768c2be1a8e4d2fcf8aa3443c7b0f084d145734c/Computability/BusyBeaver/Research/PA.

[2] Supplied `pa194_conpa_2026-11-13.zip`, 67-reader/seven-fold primary and A/D companion; and `pa194_conpa_2026-11-13(2).zip`, 68-reader/eight-fold primary. Both are in `docs/incoming` at the pinned revision. Exact archive Git blob identifiers and table SHA-256 values are in the comparison data.

[3] Supplied `PA195_ConPA_research_2026-11-14.zip`, PA195S and PA195C constructions. Archive SHA-256: `f61f38211027db46e6fae171ad9c439f0e5d6078b16db15e67f774085579c873`. The complete PA195C instruction array is frozen and checked in this package.

[4] Supplied `pa195_conpa_2026-11-13.zip`, original bundled-diagonal calculus and ordinary syntactic PA calibration. Relevant proof text is preserved under `reference/logical_calibration/` and `references/`, with source hashes and provenance.

[5] James Chang, *BusyBeaver*, PA196 and PA209 snapshots imported under `jamesyc-upstream/`. The source register architecture, logical interfaces, generic runtime proofs, and existing formalization boundary are credited to those snapshots and their cited predecessors. https://github.com/jamesyc/BusyBeaver.

[6] *ProveIt*, `Computability/BusyBeaver/Research/ZFC`, especially the K215 source calibration and propositional basis documentation, at the same revision.

[7] *ProveIt*, `Logic/PrimitiveRecursiveArithmetic/ConciseIPRA`, and the conditional PA194 arithmetic endpoint in `Computability/BusyBeaver/Research/PA/pa194/ipra`, at the same revision. The stronger iPRA equivalence is not claimed here.